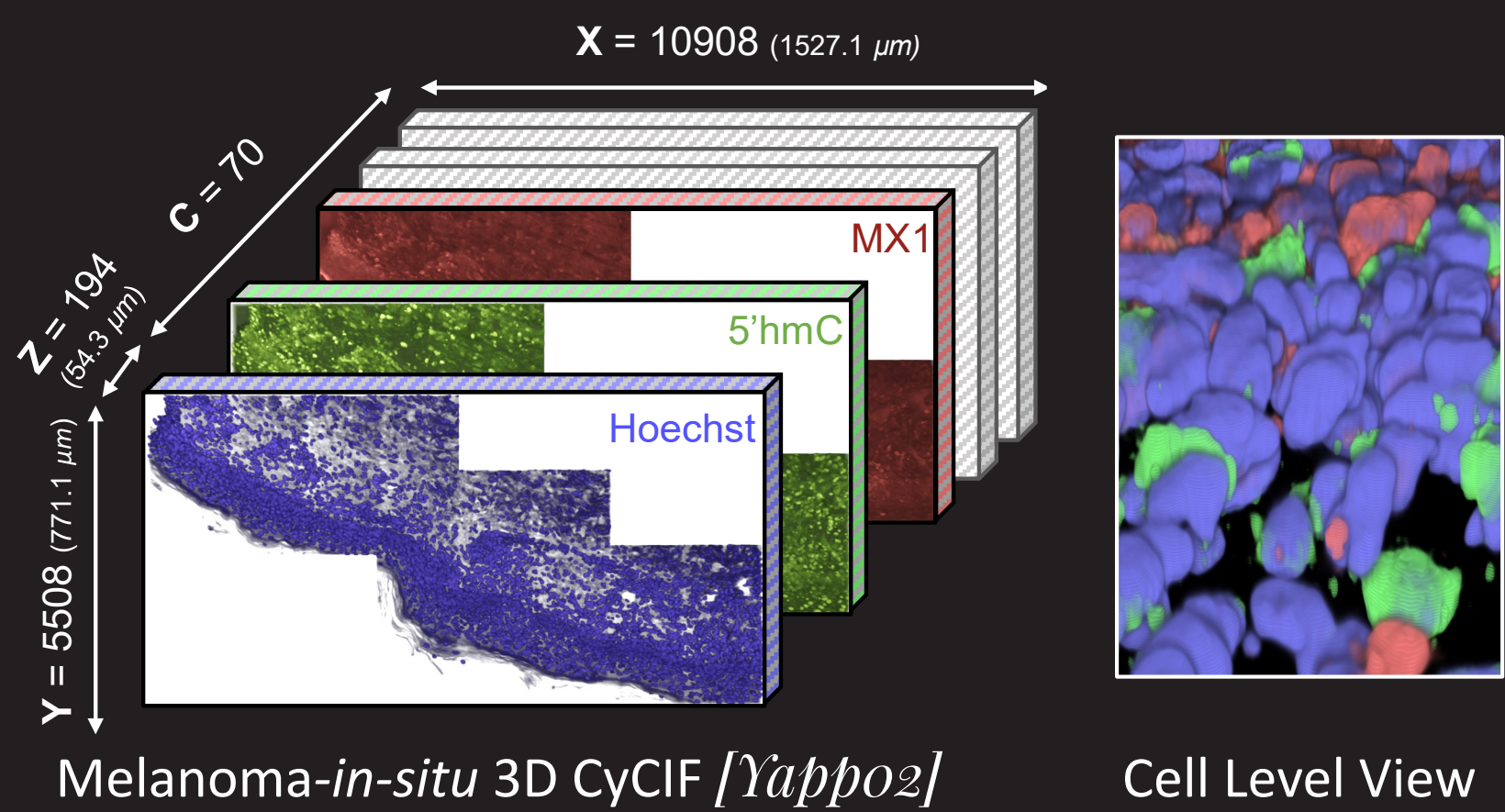


Scalable 3D Visualization and Spatial Exploration of Multiplexed Cancer Tissue Images for the Web



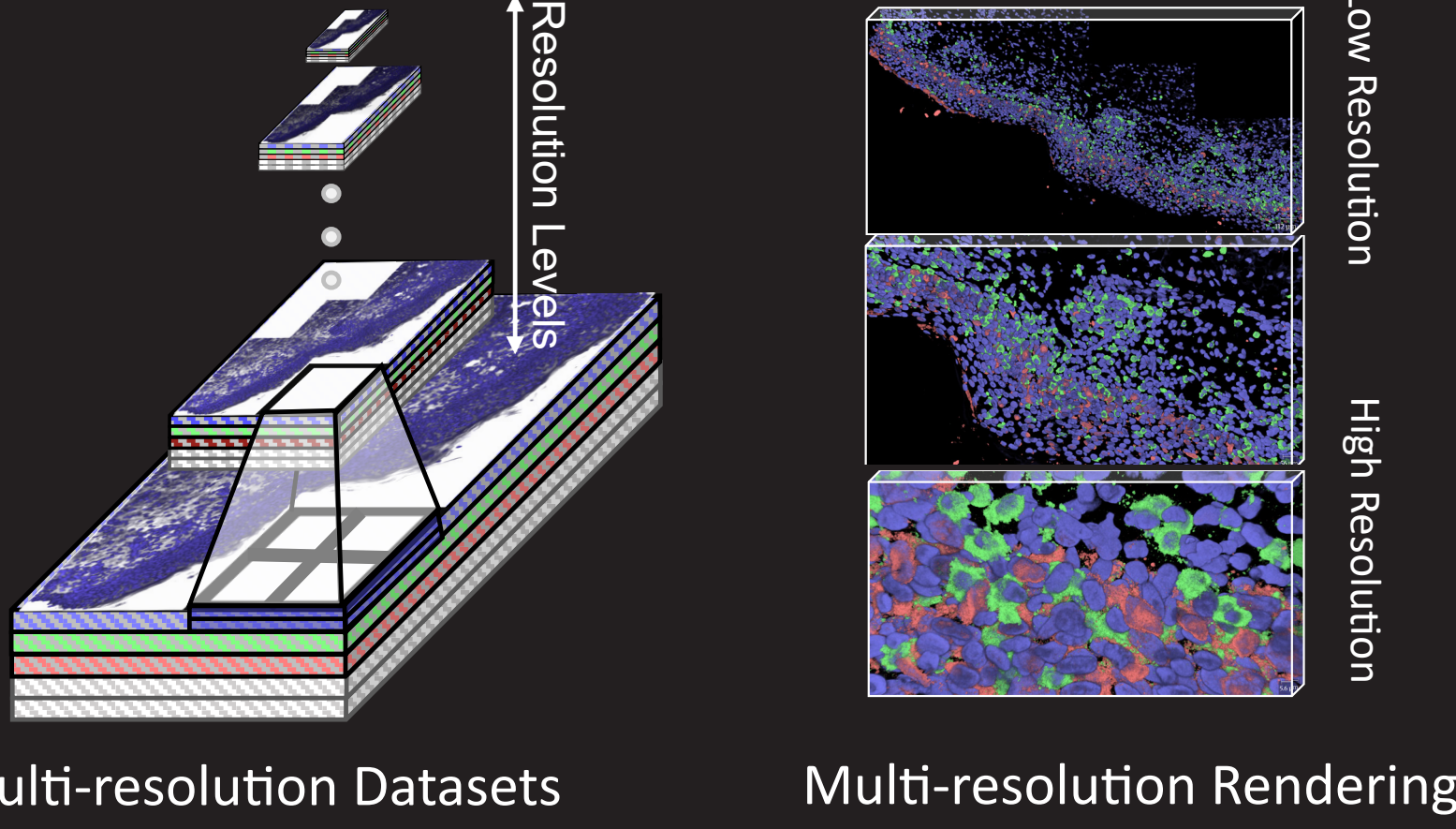
Chahat Kalsi¹, Hossein Fathollahian², Luca Reichmann³, Yuancheng Shen¹, Sophia Gaupp³, Angéla Szabó⁸, Alfredo Rodríguez⁷, Tuulia Vallius⁶, Meri Rogava⁵, G. Elisabeta Marai², Michael Krone⁴, Saeed Boorboor², and Robert Krüger¹

From 2D to 3D Cyclic Immunofluorescence



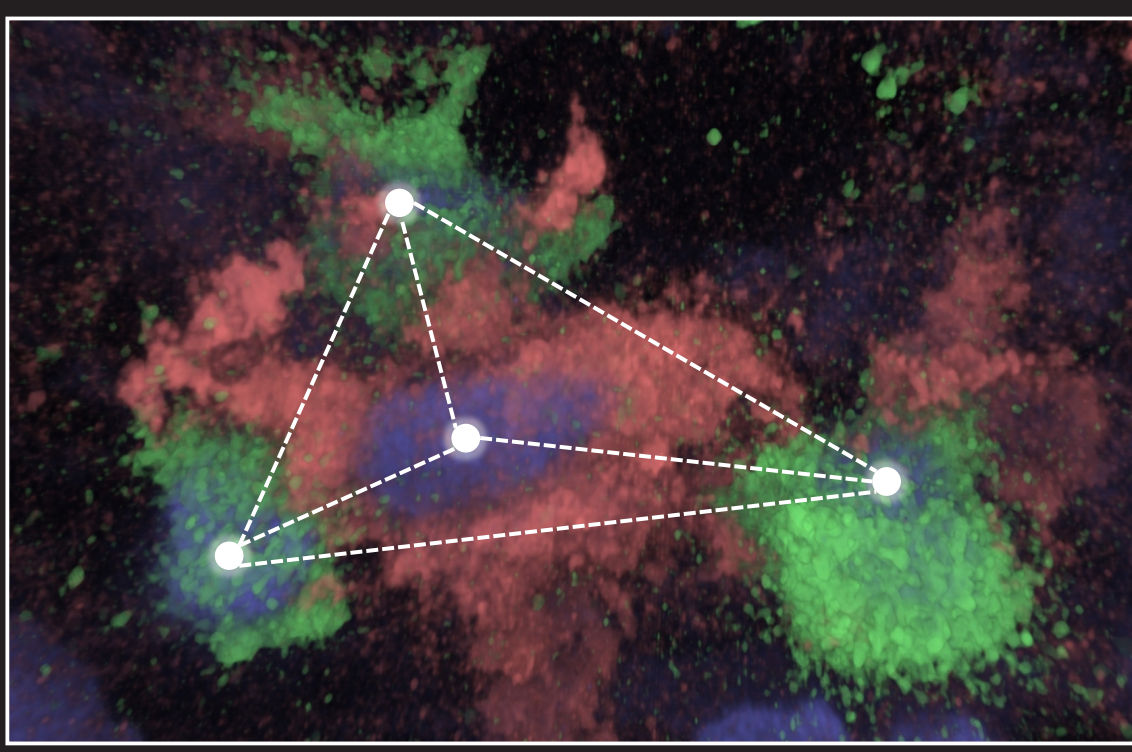
Recent3D[Yäppo2]extensionsofCyclicImmunofluorescence [Lino1] preserve more intact cells and their spatial organization than 2D imaging, but produce high-resolution, highly multiplexed, terabyte-scale datasets.

Multi-Resolution 3D Visualization for the Web



Multi-resolution formats such as OME-NGFF/Zarr enable selective loading of visible regions. Works such as Residency Octree [Herzbergero3] use these resolutions for mixed-resolution volume rendering on the web.

Visual Analytics for 3D Multiplexed Tissues



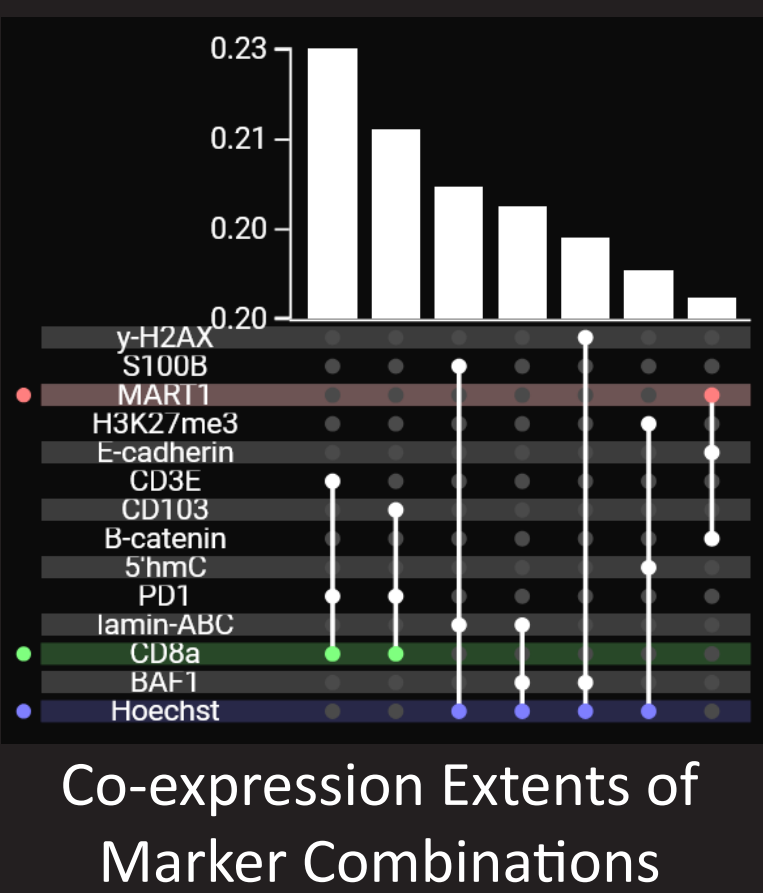
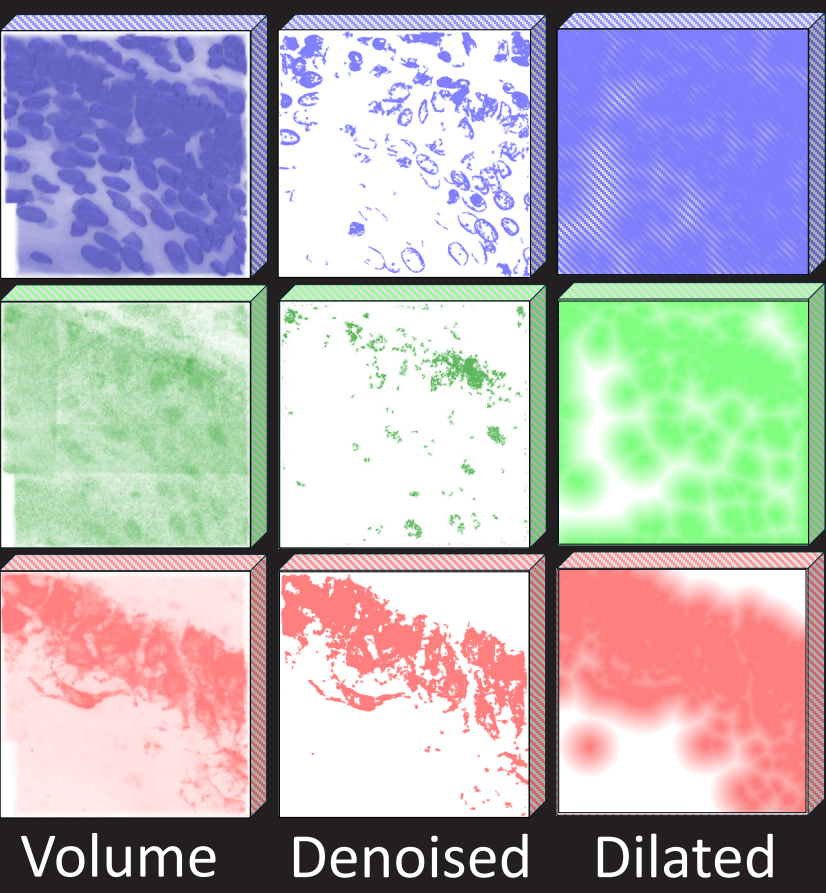
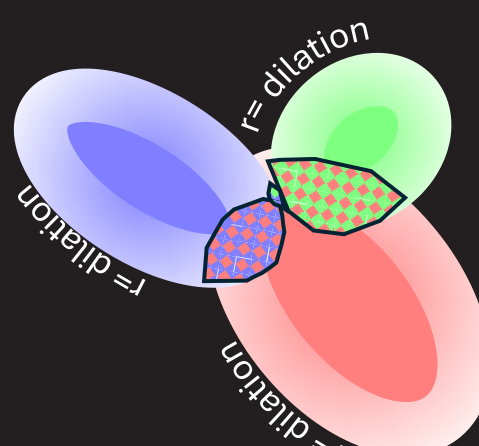
Graph-based methods [Moertho4]

Most visual analysis techniques for these 3D tissues [Moertho4] assume spherical cells and quantify relationships along edges connecting cell pairs. ConGAT [Fathollahiano5] uses a graph transformer on edges between groups of voxels.

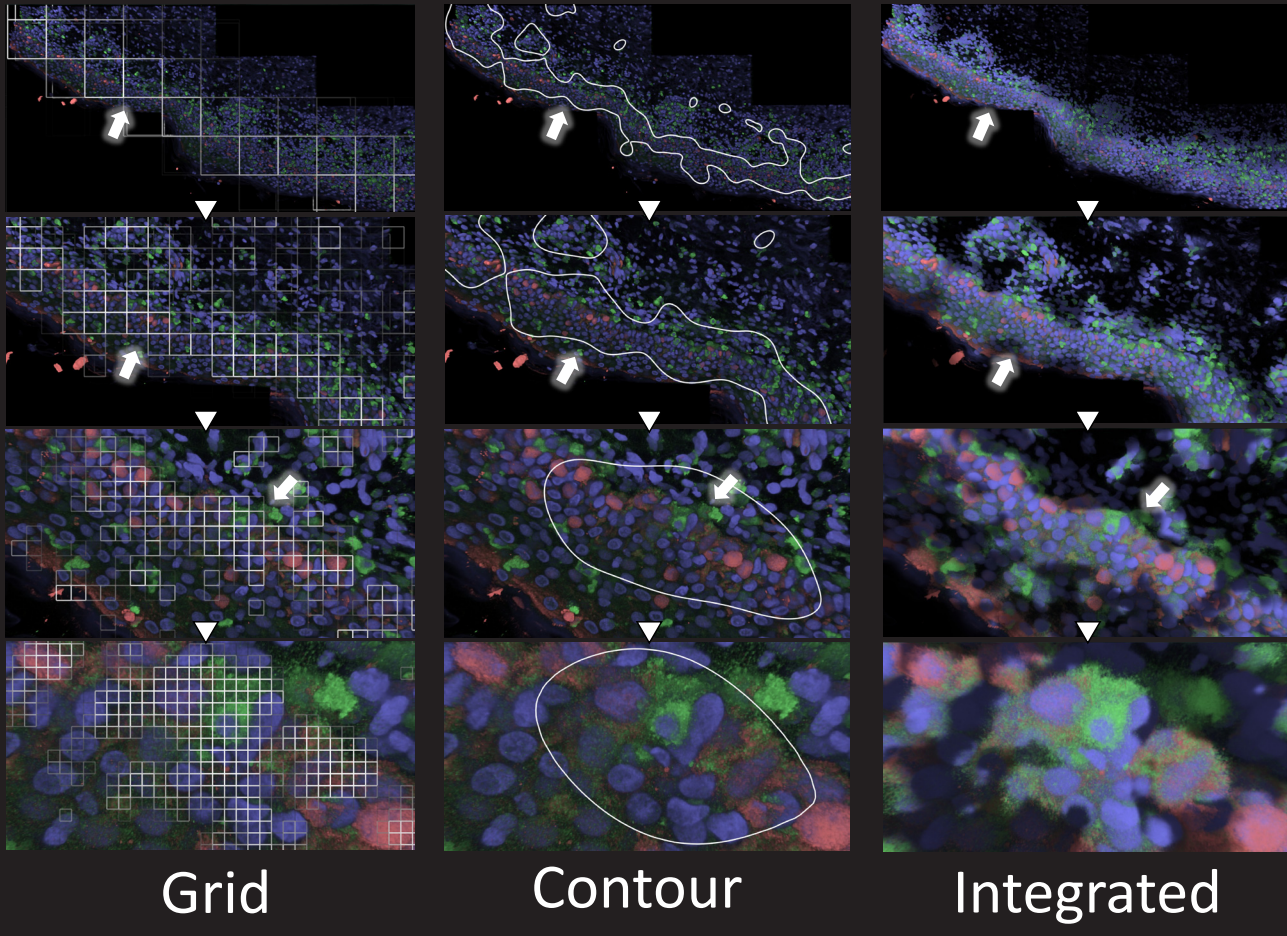
BioSET - Biomarker-based Spatial co-Expression analysis in Tumor environments

BioSET denoises channels and measures their overlap across dilations to locate and quantify spatial marker co-expression.

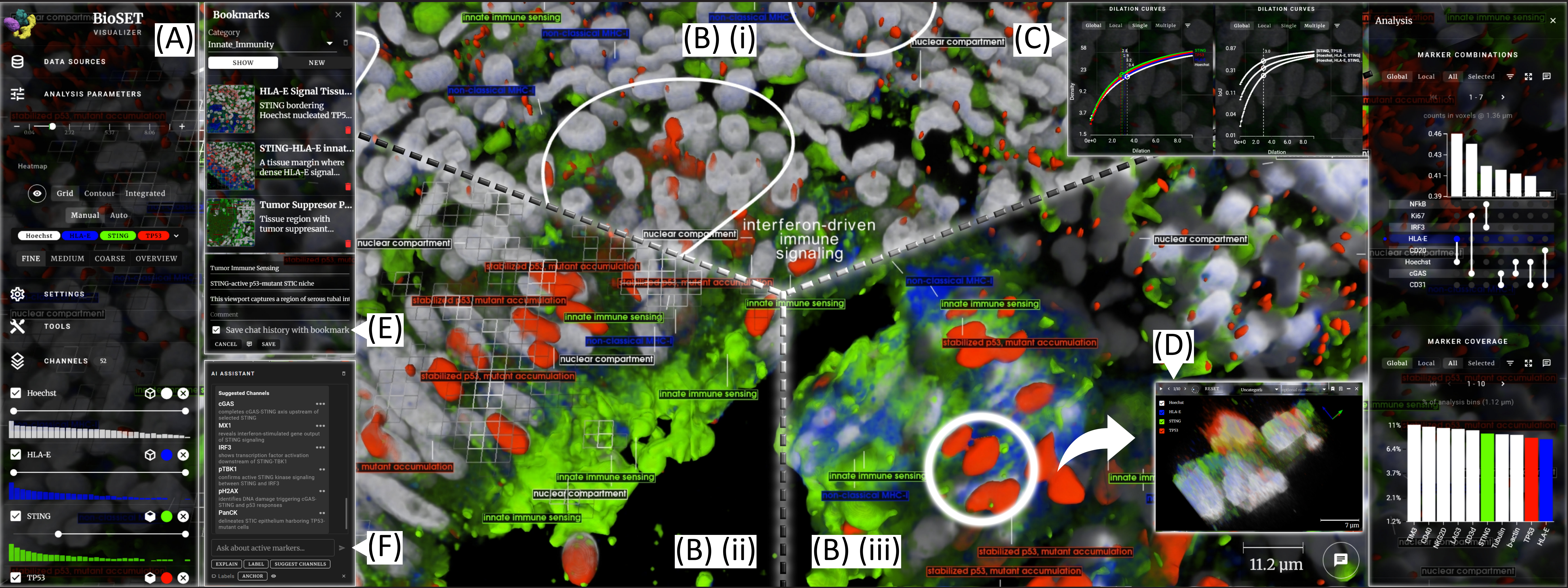
This preserves the diverse morphologies of biomarkers.



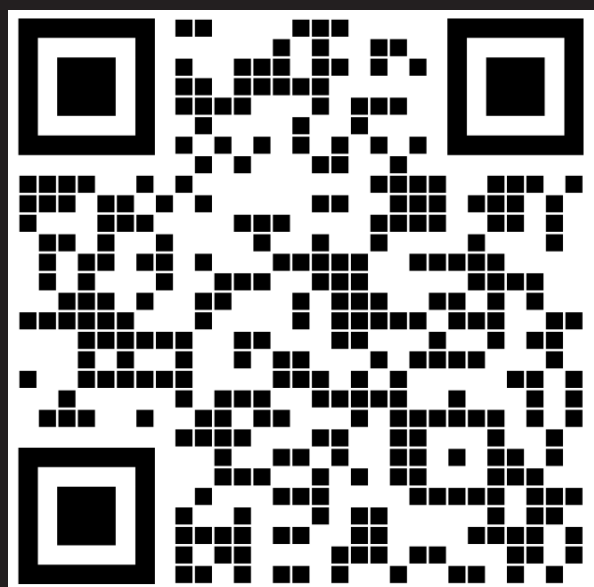
Given the highly multiplexed nature of these tissues, BioSET's plots summarize prominently co-expressed marker sets, that helps interpret these tissues. Further, embedded spatial guidance (grid/contour/integrated) helps navigate to sparse ROIs within tissues.



Evaluation: BioSET was evaluated on a 1.6 TB melanoma-in-situ [Yäppo2] dataset containing six resolution levels, through three case studies with experts, including those involved in imaging the dataset.



Overview of BioSET. (A) Sidebar for settings related to camera, viewport, data and channel selection. Intensity ranges and colors of channels can be modified. Upto 10 channels can be selected simultaneously. (B) Spatial guidance mechanisms highlighting ROIs containing selected channels using (i) contours, (ii) grid, and (iii) co-localization enhancing volume rendering (integrated), with VLM-generated labels embedded in a hybrid volume and surface rendering of 4 channels. (C) Global or viewport-local summaries of marker combinations, co-expressions, and overlap using UpSet, bar, and line plots. (D) Optimal-view for region under lens. (E) AI-assisted ROI bookmarks. (F) AI-assistance suggesting channels based on current view.



BioSET GitHub

[Lino1] - Lin, J.-R. et al. 2016. Cyclic immunofluorescence (CyclIF), a highly multiplexed method for single-cell imaging. Curr. Protoc. Chem. Biol.
[Yäppo2] - Yapp, C., et al. Highly multiplexed 3D profiling of cell states and immunocytes in human tumors. NatMethods (2025).
[Herzbergero3] - L. Herzberger et al., "Residency Octree: A Hybrid Approach for Scalable Web-Based Multi-Volume Rendering," in IEEE TVCG, 2024.
[Moertho4] - E. Mörth et al., "Cell2Cell: Explorative Cell Interaction Analysis in Multi-Volumetric Tissue Data," in IEEE TVCG, 2025.
[Fathollahiano5] - H. Fathollahian et al., "ConGAT: Context-aware graph attention network for 3D region of interest discovery in multiplexed microscopy images," in Computers & Graphics, 2026.



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